

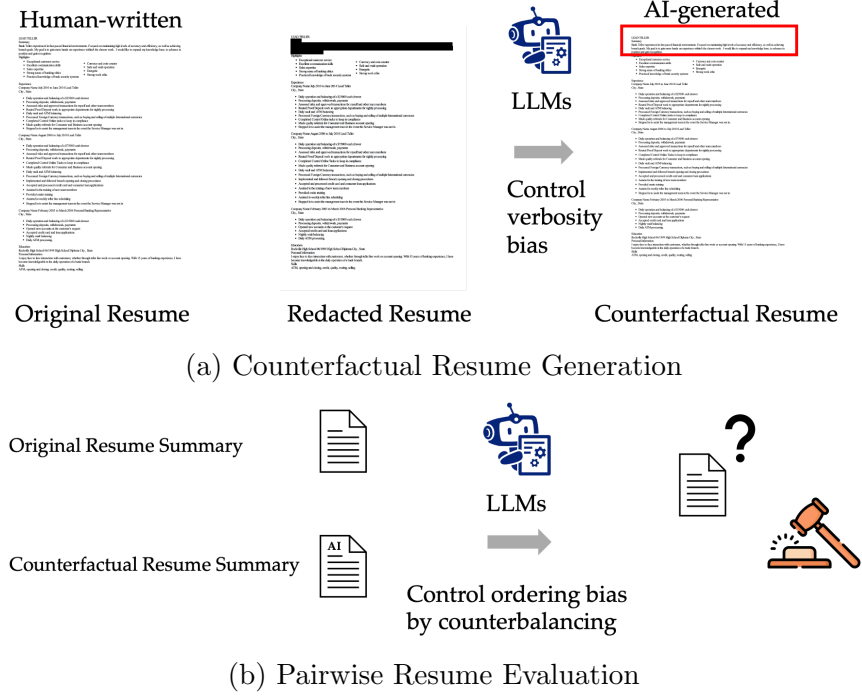


Figure 2 Resume Correspondence Experiments

4.2.2. Pairwise Resume Evaluation

To test for AI self-preference bias, we conduct pairwise evaluations based on the executive summary text. Although counterfactual summaries are generated conditional on the full resume to ensure that all versions are grounded in the same underlying candidate information, evaluator LLMs are shown only the two executive summaries when making their selection. Since all other resume content is identical across counterfactual versions, it is omitted from the evaluation stage.

Focusing the evaluation on the executive summary ensures that comparisons are made between alternative textual representations of the same candidate profile, without introducing additional variation from other resume sections. This design allows us to attribute differences in evaluation outcomes to the source and expression of the summary text, rather than to differences in underlying candidate information or document-level alignment.

To simulate a resume screening task, we designate one LLM as the evaluator LLM and refer to all other models as alternative LLMs. As illustrated in Figure 2b, the evaluator LLM is prompted to compare two executive summaries and select the stronger one based on the overall presentation of skills and experience, following a standardized, job-agnostic evaluation instruction. For each evaluator LLM, we construct two types of summary pairs: (1) a summary generated by the evaluator LLM versus a human-written summary, and (2) a summary generated by the evaluator LLM versus one generated by an alternative LLM. The full prompt for the evaluation task is provided in Appendix B.

To account for position (or ordering) bias (Zheng et al. 2023), where LLMs may exhibit a preference for the first or second option presented, we implement a counterbalanced design following Brooks (2012). Specifically, the order of resumes (i.e., A vs. B) is randomized across comparisons. This approach ensures that any position-related biases are evenly distributed and do not systematically affect evaluation outcomes, thereby preserving internal validity without requiring duplicated evaluations. See Appendix C for examples of both human-written and LLM-generated professional summaries used in the pairwise comparisons.

5. Empirical Results

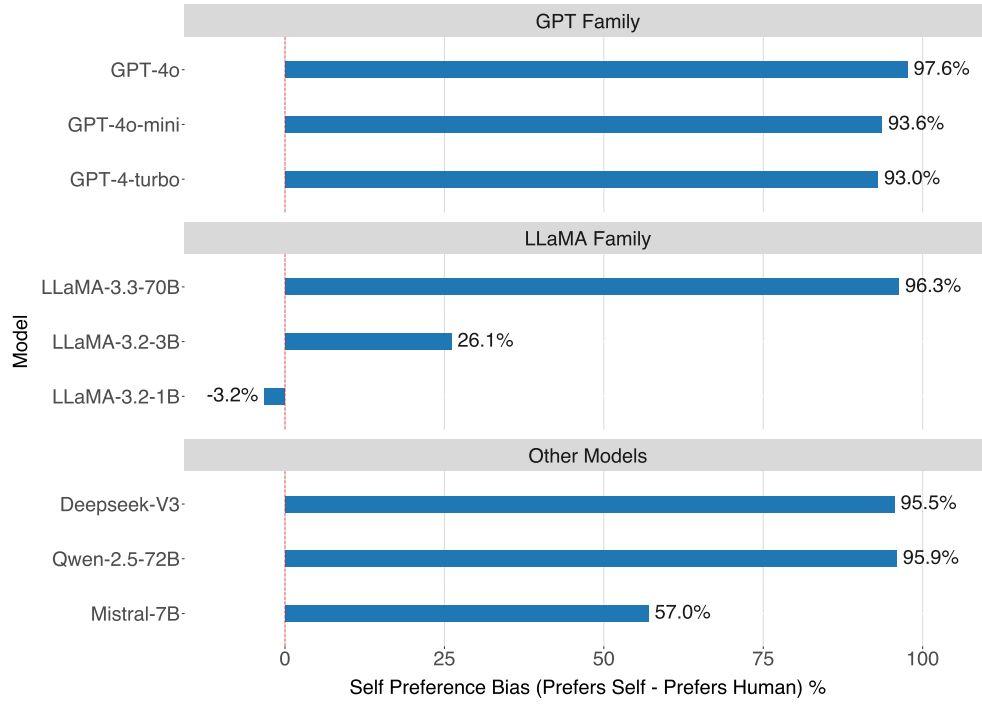
Building on the definition and measurements of AI self-preference bias introduced in Section 3, we empirically evaluate the extent to which LLMs favor their own generated content over that produced by humans or alternative LLMs. We begin by analyzing the LLM-vs-Human self-preference, followed by the LLM-vs-LLM self-preference. For each form of bias, we present results based on the two fairness metrics (statistical parity and equal opportunity) to provide complementary perspectives on self-preferencing behavior.

5.1. LLM-vs-Human Self-Preference

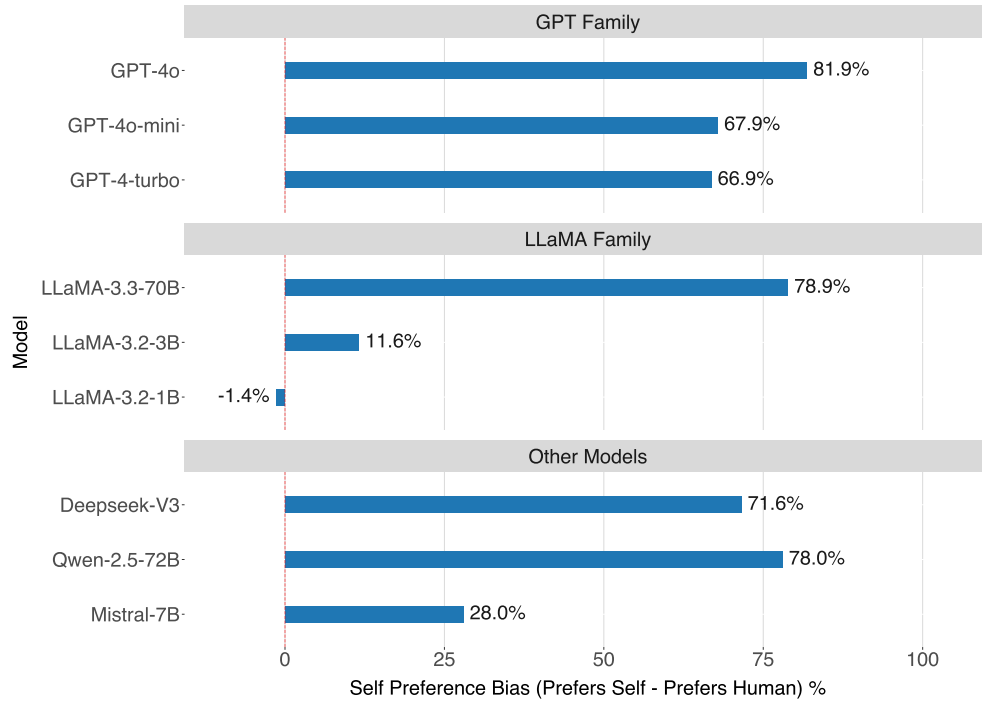
We find strong and consistent evidence that LLMs, when used as evaluators, systematically prefer resumes they generated themselves over equivalent resumes written by humans. This bias is pervasive across models and persists even after controlling for content quality. Notably, the strength of self-preference increases with model size, which may indicate that larger models are more sensitive to stylistic features resembling their own outputs.

5.1.1. Statistical Parity Self-Preference Bias

Under the statistical parity metric defined in Eq. (1), we compare the probability that an evaluator LLM selects its own generated resume with the probability that it selects a human-written resume. The difference between these probabilities provides a direct measure of self-preference bias: positive values indicate systematic favoritism toward the model’s own outputs, values near zero reflect parity, and negative values indicate a preference for human-written resumes. As shown in Figure 3a, each bar reports the outcome of an evaluator LLM choosing between a human-written resume and a version it generated itself. Eight of the nine LLMs tested exhibit clear LLM-vs-Human self-preference, with magnitudes ranging from 26% to 98%. In practical terms, this means evaluator LLMs are between 26% and 98% more likely to select a resume they generated themselves than an equivalent resume written by a human.



(a) LLM-vs-Human Statistical Parity Self-preference Bias



(b) LLM-vs-Human Equal Opportunity Self-preference Bias

Figure 3 LLM-vs-Human Self-Preference Bias

5.1.2. Equal Opportunity Self-Preference Bias